

Enterprise Digital Infrastructure – Self-assessment: DNS cache

April 14, 2026

First name/Lastname _____

ID _____

1. Perform the queries to obtain the IP addresses of the following hostnames. For each query specify the Names Servers involved and the content of the various caches and compute the number of RTTs required to obtain each IP address.

Assumptions: the Local Name Server is the official Name Server of the domain **unipv.it**. Caching is enabled and all caches are initially empty. The TTL associated with TLD Resource Records is set to 600 seconds, while the TTL of second level domain Resource Records is set to 300 seconds.

a. At time $t=0s$ **www.webapp.io**

b. At time $t=10s$ **ercole.unipv.it**

c. At time $t=100s$ **www.abc.io**

d. At time $t=200s$ **ercole.unipv.it**

e. At time $t=300s$ **www.abc.org** [the hostname does not exist]

f. At time $t=400s$ **www.webapp.io**

g. At time $t=500s$ **www.abc.org** [the hostname does not exist]

h. At time $t=1000s$ **www.unipv.it**

2. Consider a scenario where the Local Name Server is a public recursive resolver (e.g., 8.8.8.8 or 1.1.1.1) and discuss the differences (if any) in the RTTs spent for performing the queries of Exercise 1. Use the same assumptions of Exercise 1.

a. At time $t=0s$ **www.webapp.io**

b. At time $t=10s$ **ercole.unipv.it**

c. At time $t=100s$ **www.abc.io**

d. At time $t=200s$ **ercole.unipv.it**

e. At time $t=300s$ **www.abc.org** [the hostname does not exist]

f. At time $t=400s$ **www.webapp.io**

g. At time $t=500s$ **www.abc.org** [the hostname does not exist]

h. At time $t=1000s$ **www.unipv.it**